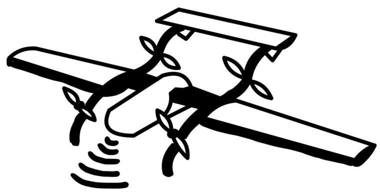
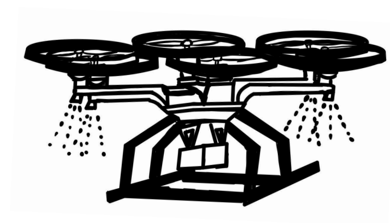
TITAN: Terrain Imaging for Targeted Application of Nanofertilizers





Columbia University Space Initiative

Gateway to Blue Skies Competition 2025

Ashika Gullapalli, Ahmed Jaber, Maxim Koretsky, Gev Nweke, Victoria Sepp, and Noah Thomas

Team Members from left to right:
Ahmed Jaber, *Freshman*, Computer Science
Maxim Koretsky, *Junior*, Computer Science
Victoria Sepp, *Freshman*, Electrical Engineering
Noah Thomas, *Junior*, Mechanical Engineering
Gev Nweke, *Junior*, Mechanical Engineering
Ashika Gullapalli, *Sophomore*, Environmental Biology
Jose Osa, *Junior,* Materials Science Engineering

Team Lead and Primary Contact: Maxim Koretsky
Email: mk4990@columbia.edu Phone: (408) 329-8610

Team Lead and Primary Contact: Noah Thomas
Email: nst2121@columbia.edu Phone: (617) 372-2973

Team Advisor: Astronaut Mike Massimino
Professor of Professional Practice
Mechanical Engineering, *Columbia University*

Email: mmassimino@columbia.edu
Phone: (212) 854-3304
Advisor Signature of Review and Approval:

Additional Thanks to:

Dr Walter Baethgen, Senior Research Scientist at Columbia Climate School
Professor Gerald Holmes, Ph.D, Strawberry Center at Cal Poly, San Luis Obispo
Professor Glenn Denning, Director of MPA Development Practice at Columbia University
Matt Haberhand, Assistant Professor, BioResource and Agricultural Engineering Department
Ochs Orchard, Warwick, NY
Shannon Riley, Agricultural Sciences, Oregon State University

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Abstract

Traditional fertilizer use is one of the top contributors to the climate crisis today [49]. Accounting for over 5% of greenhouse gas emissions, the production and application of synthetic fertilizers are detrimental to our environment and health [50]. Current fertilizer application methods are wasteful, and modern technological developments are often too expensive or have too many technological barriers for farmers to adopt themselves. To remedy this, we propose TITAN, an autonomous aerial system of one high-speed mapping drone and one nanofertilizer deploying hexacopter drone designed to reduce fertilizer use by over 80%. TITAN targets medium to large farms, where farmers cannot easily track and assess their crops without automation. The AURAS mapping drone is an autonomous, lightweight, fast-moving drone that can cover five to ten times more farmland than a traditional drone in a single flight. Since AURAS is an autonomous drone, it has a low barrier of entry as farmers do not need to learn how to operate the drones. This mapping is paired with the deployment of nanofertilizers via our CLOUD hexacopter drone. What sets nanofertilizers aside from traditional fertilizers is their higher surface area allowing them to be more efficiently absorbed by crops, slow-releasing capabilities causing them to nourish crops for longer, and increased nutrient density to increase crop yield while lowering fertilizer volume. Our system is designed to 1) lower the barrier to entry for sustainable fertilizer technologies. 2) Employ the slow-releasing and precision targeting technologies of nanofertilizers to limit runoff and fertilizer volume. 3) Increase nanofertilizer research and development to increase their availability and market potential. Drone development and nanofertilizer research have been promising in recent years. However, there is still a need for research on the long-term effects of nanofertilizers through multi-year field trials in the coming years. Large-scale testing must be conducted on commercial farms of varying sizes on various crops. Lastly, machine learning processes for mapping drones must be conducted and nanofertilizer production must increase for TITAN to be affordable and adopted by farms across the country. We anticipate full implementation of the system by 2035. We forecast that TITAN will save each farm $93,000 per year with only a $50,000 one-time installation payment. The TITAN system also is estimated to reduce US CO_2 emissions by 3.6%, or 230,000,000 metric tons.

Quad Chart

TITAN

Terrain Imaging for Targeted Application of Nanofertillizers

GATEWAYS TO BLUESKIES

Project Summary :

Fertilizer application is a costly and environmentally damaging process. Over 5% of global carbon emissions come from fertilizer production and application, with the latter accounting for two-thirds of these emissions. **TITAN** is a mapping and fertilization system designed to eradicate emissions in the application process. The system combines GPS naviation and a custom R-CNN/ViT classifier model with the precise application of nanofertilizers. Nanofertilizer particles can be absorbed by crops and soil more efficiently than synthetics, limiting fertilizer use and pollution while improving crop yield.

Project Image:

Team Composition/Roles:

We are a diverse club at the **Columbia Space Initiative** composed of undergraduates of various grades and majors. After interviewing farmers and experts in the agriculture industry, we are proposing a system that will initiate collaboration of industry experts, scientists, and engineers to support sustainability, public health, and economic growth.

***Ashika Gullapalli**, Environmental Strategist*
Sophomore, Environmental Biology
***Ahmed Jaber**, UAV Specialist*
Freshman, Computer Science
***Maxim Korestsky**, Managing Director*
Junior, Computer Science
***Gev Nweke**, Mechanical Expert*
Junior, Mechanical Engineering
***Victoria Sepp**, Systems Analyst*
Freshman, Electrical Engineering
***Noah Thomas**, Operations Director*
Junior, Mechanical Engineering

Proposed Deployment Timeline:

Nanofertilizer Research	*Prototyping and Testing*	*Field Trials and Machine Learning*	*Integration*
2025 – 2028	2028 – 2030	2030 – 2032	2032 – 2035

2025-2028: Investment in research of long-term nanofertilizer impact.

2028-2030: Establishing nanofertilizer regulatory standards. Prototyping AURAS and CLOUD drones.

2030-2032: Large scale field tirals with pilot programs and data collection for ML model training.

2032-2035: Upscaling nanofertilizer production and full integration of TITAN.

I. <u>Current State Assessment</u>

As the US population continues to grow, so does our need for agricultural production. Every year, farmers are faced with the challenge of keeping up with this demand. In recent years, farmers have met this challenge by dispersing 22 million tons of liquid fertilizer across US farmland every year [18]. Without fertilizer use, corn and wheat plants in the US would lose 60% and 40% of their annual yield, respectively [17]. This fertilizer use is essential, but it is also a dated process that has several drawbacks.

A. Environmental and Health Concerns of Fertilizers

Nitrogen-based fertilizer use is over 40 times higher than it was 75 years ago, and nearly two-thirds of all nitrogen applied to crops becomes a pollutant [1]. Excess fertilizer use contributes to climate change, hypoxic dead zones, and toxic algal blooms [4]. Nitrous oxide released during the fertilizer application process has a global warming potential 250–300 times greater than that of carbon dioxide, and the use of fertilizers accounts for 5.1% of total global greenhouse gas emissions.

Beyond its impact on the environment, inefficient fertilizer use contributes to human health concerns as well. A 2019 study linked 4,300 premature deaths in Corn Belt states to nitrogen fertilizer use and their release of ammonia [1]. Recent findings from the EPA also highlight that many fertilizers in use today contain per- and polyfluoroalkyl substances, both of which are linked to cancers and other serious health concerns [19]. Fertilizers are contaminating our atmosphere and our water. If this trend of increased fertilizer use continues without intervention, the health of our planet and our people is at risk.

B. Economic Drawbacks

In addition to this impact on our planet and health, fertilizers pose an annual cost exceeding 157 billion dollars to farmers, fishermen, and the government. For instance, just one algal bloom in Maine caused by nutrient pollutants cost the fishing industry over $3 million, while another persistent bloom in Ohio cost the state over $40 million in lost tourism revenue over two years. Additionally, fertilizer production costs are directly related to energy costs. When the Russia-Ukraine war drove energy prices to rise in 2022, the cost of fertilizer production nearly doubled from $483 per metric ton to $925 [20]. The current administration poses several complications to labor costs as well, as an estimated 42% of hired farm workers in the US have no work authorization [21].

Fertilizer-deploying machinery adds to this cost. In Indiana, rates per acre increased 37% since 2017 and 27% since 2021, due to the increased cost of parts and labor following the pandemic [60]. The world's largest farm equipment company, John Deere, reported a revenue decrease of 35% in their latest quarterly report [61]. Deere reported that prices for industrial metals, critical components for Deere's products, have dramatically increased due to tariffs on steel and aluminum. TITAN would help alleviate such financial pressures for farmers with new technology aimed at decreasing costs.

C. Summary of Current Technology

Despite technological advancements that aim to limit fertilizer use and its environmental impact, fertilizer use remains a cornerstone of United States agriculture. Synthetic fertilizers require the use of fossil fuels to be produced, pollute the atmosphere when deployed, leach into and poison water sources, run off into rivers and oceans, create hypoxic dead zones, degrade our soil, and contribute to biodiversity loss. As the United States grapples with these issues on the backend, an estimated 60% of applied fertilizer is never used and acts entirely as a pollutant [27].

Today, application methods of fertilizers include spreading fertilizers uniformly across fields, applying fertilizer in rows, and placing fertilizer at the bottom of furrows. These application methods all lead to air or water pollution, and while technological advancements aim to mitigate these damages, few are cost-effective, making it difficult for small farmers to adopt a new system.

Due to a lack of access to technology and machinery, small farms use more fertilizer per hectare than large farms. Moreover, large farms dominate the fertilizer market and have a much higher potential for longer-term investment from government subsidies or crop insurance [24]. As such, sustainable initiatives targeting large-scale farms can be more easily adopted and have a higher potential environmental impact than those designed for small farms.

II. Use Case and Proposed Solution

A. Problem Statement and Identified Need for System

Current fertilizer application methods are inefficient, wasteful, and cause irreversible environmental damage [55, 56, 57]. Broadcasting across acres of farmland is the most common application method in farms today, but it is far from the most efficient. This broad dispersion of fertilizer accounts for 312.6 MMT and 274.8 MMT of nitrous oxide and methane gas emissions nationwide, and reducing fertilizer use by just 10% reduces methane emissions by up to 22.11% [52].

Several modern agricultural technologies aim to improve fertilization efficiency, yet each has its limitations. Shannon Riley, a vegetable farmer in California and a graduate of Oregon State University, explained, “Being sustainable is a plus for farmers…but they only invest in new tech if they’re confident it’ll save them money long-term” [22]. Many technologies are yet to get past this hurdle.

It is important to note that the agriculture sector is experiencing a labor shortage of farm workers, predicted to worsen, due to the current administration’s policy of mass deportations targeting individuals from the southern border. Current estimates have 60-70% of the farming workforce composed of latino farm workers [58]. Furthermore, during the pandemic, the percentage of farms reporting labor shortages was roughly 53% [33]. TITAN’s autonomous features will aid farmers to better allocate their limited resources, especially in consideration to labor shortages.

B. Emerging Technologies

1. Mapping with Multispectral Cameras/Sensors’

Multispectral imaging in agriculture uses various light wavelengths to create images of one’s field. These images can detect variations in crops before visible symptoms appear. Examples of state-of-the-art technologies for mapping with multispectral cameras are the MiDAR (Multispectral imaging, Detection, and Active Reflectance) system and the THETA (Tens-of–megapixel Handheld Multi-Spectral Videography Approach). The key point of MiDAR is the adaptive sensing feature, where the system can change color bands and relative intensities in real-time. MiDAR’s advancements have a lower signal-to-noise ratio, which in contrast to outputting all wavelengths simultaneously outputs clearer data and is more energy efficient and critical for mass and power systems such as drones. This technology represents a significant upgrade from previous approaches that only measured at predetermined spectral bands. With MiDAR, the system can respond in a much quicker fashion to various altering conditions during data collection [35, 36]. The THETA system represents a breakthrough in multispectral imaging technology, achieving up to a 65-megapixel video resolution with 12 wavebands within the light range, one of the highest resolution systems currently available. In addition to massive data collection, the THETA system has a deep neural network-based architecture tailored for multi-spectral data encodings [37]. These systems can detect NIR values reflected by plants, which can be used to calculate vegetation indexes (eg.. NDVI) [38]. These are then fed into the ML model to predict plant health.

2. Nanofertilizer Development

Nanofertilizers are made by synthesizing nanoparticles such as carbon nanotubes, graphene, and metal oxides [51]. These particles have unique properties such as high surface area to volume ratios, biocompatibility, and controlled release mechanisms. By delivering nutrients to plant roots directly, they can penetrate plant tissues more effectively in comparison to synthetic fertilizers. Moreover, through nanofertilizer application, plants obtain an enhanced ability to withstand abiotic stress [66]. While nanofertilizers are more expensive per unit in contrast to traditional fertilizers, the cost is offset by increased efficiency and crop yield. For example, nanofertilizers have increased the crop yields of corn, wheat, and millet, leading to an increase in revenue of $133.2, $86.4, and $103.1 per hectare, respectively [23]. It is important to consider the decrease in volume with nanofertilizer use, as one field study demonstrated that nanofertilizers require one-fifth of the volume to achieve the same yields as a traditionally fertilized plot [23]. Especially when considering the environmental costs ranging from $500 to $1,000 per hectare from traditional fertilizers, sustainable nanofertilizer use becomes the most economically beneficial option among emerging technologies [20].

3. Mapping Drones

Existing drone technology similar to the proposed mapping system is the "FIXAR 007" [39]. The current specs of the FIXAR 007 include a range of 37.3 miles, cruise speed of 38.03 MPH, flight time of roughly an hour, payload of up to 4.5 pounds, vertical take-off and landing, proprietary autonomous operation software, fixed angle rotors, and information gathering capability of 6,000 HA at 5 cm/px. Other options for mapping drones include the DJI Matrice 300 RTK [40]. The payload capacity, flight time, and sensor compatibility are quite similar to the FIXAR 007, but the DJI Matrice 300 RTK has a high weather resistance (IP54 on the FIXAR compared to IP45 on the 300 RTK) as well as an advanced automation feature. However, a key downside of the 300 RTK is the operational range is only up to 15 km (9.3 miles) for HD transmission, which is quite low.

4. Fertilizer-Deploying Drones

Aerial fertilizer or pesticide spraying vehicles are generally a new advent in the agriculture sector with "the global agriculture drone market projected to grow from $4.98 billion in 2023 to $18.22 billion by 2030, at a compound annual growth rate of 20.3% during the forecast period." [41] This technology initially started picking up traction in East Asia and has slowly emerged in the US market. A Chinese company has gained attention in the US using a drone called the "DJI AGRAS T50". This drone is extremely advanced but has some major downsides including a very limited flight control range as well as relatively slow max speed. It has also been reported by users that the mechanical components of the drone are not robust enough and that it has major overheating issues and compass malfunctions. [42].

III. Implementation

A. Operations and System Overview

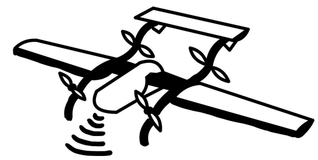
The field operation of TITAN will occur in two phases. First, an aerial surveying drone will conduct scanning of farmland and provide a detailed map of nutrient composition in soil. In the second phase, a low-flying drone equipped with a pressurized liquid delivery system and nan fertilizers will deliver nutrients to detected areas of nutrient deficiency. Dividing this system into two aerial components allows for the mapping drone and deployment drone to fly at different heights and speeds without having to carry any extra weight to maximize flight time and efficiency.

After initially deploying nanofertilizers across farmland to promote early crop health before planting, TITAN is designed primarily to monitor soil composition across seasons to reinforce crop health during growth. This design will replace the current system of fertilizer application, where fertilizers are wastefully sprayed in bulk across acres of farmland, with no consideration of precision or conservation.

As part of TITAN's commitment to energy conservation, farmers have the option to adopt solar-powered charging stations depending on the solar-powered capabilities of their farm location. The future of TITAN is to use renewable energy and efficient AI-driven flight patterns to increase lifespan and decrease operating costs. We can see current examples of this in use with Amazon's MK30 drone. [62]

B. Imaging Drone (AURAS)

1. Proposed System

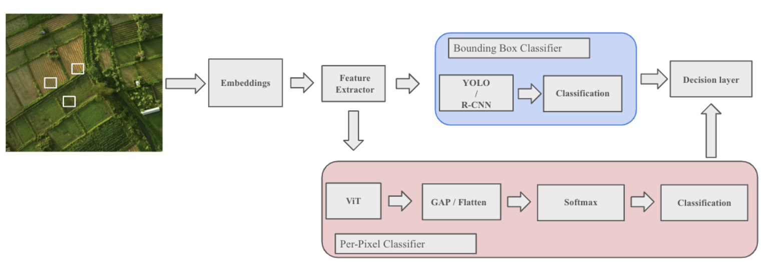


We propose the All-in-one Uniform Remote Aerial System (AURAS), which conducts aerial surveying via Normalized Difference Vegetation Index (NDVI) processing of hyperspectral data and multispectral imaging to create a nutrient composition map.

a. Specifications and Initialization

The drone begins with predefined starting GPS coordinates (Lat_0, Long_0, Alt_0) using geospatial data (GEOJson). The field is pre-processed and divided into a uniform grid of georeferenced waypoints. AURAS uses a reinforcement learning-based model to create an adaptive flight path to minimize flight time while maximizing survey space. The system also processes the multispectral data using an aggregate overlay of NDVI analysis data, thermal imaging data, and soil reflectance data to classify crop health. The ML model generates a high-resolution prescription map for the CLOUD drone, assigning optimal fertilizer application rates to different zones based on plant needs. At each waypoint, it captures high-resolution multispectral and RGB images, embedding GPS data into metadata. These images overlap by at least 70% (forward and side overlap) to ensure seamless ortho mosaic reconstruction [43]. Our imaging process will use advancements in machine learning and systems integration based on current research in software engineering, microchips, and data processing. AURAS collects an array of aerial images from embedded GPS data to ensure full coverage and precise geolocation. Using Pix4Dmapper, the images can be stitched together to create a single, spatially accurate map with each pixel having accurate coordinate metadata [43].

b. Support System Design



Following the creation of this map, the field gets divided into patches and is inputted into the embeddings layer (16x16 pixels each) [43]. Next, the data goes through the feature extraction layer. The data is then classified as a dual branch with two different classifiers. The blue bounding box classifier processes data using R-CNN or YOLO to detect high-density soil malnutrition areas. The bounding box of a 16 by 16 pixel patch in photos is roughly equally 432.64 square centimeters on the ground, assuming typical mapping conditions. This calculation was done using a common camera for NDVI analysis: the Foxtech Map-A7R II. At approximately 100-meters above ground, the ground sample distance (GSD) would be roughly 1.3 cm/pixel therefore the box size would (16 pixels * 1.3 cm/pixel) squared or 432.64

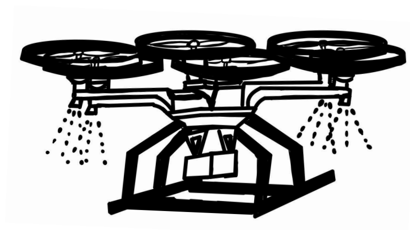
square centimeters. The red box classifies pixels individually using a Vision Transformer (ViT) for segmentation. The output of ViT is summarized into a single value for each pixel. Lastly, the softmax layer transforms the output into a probabilistic distribution with the probability being values between 0 through 1. For example, a value of 0.63 would mean the area has a probability of being 63% healthy. Both classifiers run in parallel and produce a detailed health assessment of the plot of land. The system refines spray application by detecting nutrient-deficient areas with pixel-level accuracy.

c. Projected Improvements

Given the pace of technological advancements, AURAS could be taken a step further to combat current issues such as computing time and processing time. In the future, AURAS could learn from raw, unlabeled data, such as crop yield data, as opposed to preprocessing large amounts of labeled datasets, using self-supervised learning. Rumors have Apple's M5 chip to contain 20 billion transistors and using Moore's law, current estimates for 2035 are around 600 billion transistors. AURAS could potentially use this increased computing strength to train and fine-tune the models at a fraction of the compute cost. This will make way for live-time analysis. Lastly, AURAS hopes to enhance its network connectivity with future 6G data streaming for reduced latency restrictions and increased distance of connectivity.

C. Deployment Drone (CLOUD)

1. Proposed System



The Controlled Low-Altitude Optimized Uniform Deployment (CLOUD) Drone targets nutrient-deficient zones using the mapping data from AURAS. The low-flying CLOUD drone operates autonomously. CLOUD spray liquid nanofertilizers onto fields with an AI-driven navigation system. The fertilizers will be naturally slow-released and absorbed by crops for 40 to 60 days [31].

a. Consideration of Environmental Conditions

We considered several environmental conditions when designing the CLOUD drone, specifically those of a low-flying drone designed to spray nanofertilizers. The hexacopter's six rotors stabilize the drone better than a standard quadcopter, which is critical to maximize precision when applying nanofertilizers [32]. The frame will be constructed from weather-resistant and lightweight carbon fiber to withstand various environmental conditions. Accounting for changing wind and weather patterns, the CLOUD drone is equipped with anemometer sensors and dynamic flight systems, continuously adjusting all six rotors in real time to fly as smoothly as possible. [34]. The aerodynamic performance and structural integrity of the drone have been validated through rigorous testing and evaluation, ensuring it can sustain operations in weather conditions such as strong winds, high humidity, and moderate precipitation [65].

b. Nanofertilizer Deployment Support System

The nanofertilizer deployment system features a pressurized delivery system to control output volume and limit both excess fertilizer use and malnutrition during application. Additionally, CLOUD uses electrostatic spraying technology, a method of charging particles before deployment. Electrostatically charging particles ensure nanofertilizer particles adhere to crops, further limiting runoff and overuse.

CLOUD uses LiDAR and ultrasonic sensors to maintain optimal height above crops using terrain-following capability. CLOUD further limits flight time, energy consumption, and nanofertilizer use by following an AI-optimized flight path.

Lastly, CLOUD's fertilizer reservoir consists of an auto-refill docking system at the charging station. The system mimics a lamp post with the charging and refill on the top of the post. This aims to improve autonomy and limit human intervention for maintenance. The reservoir also optimizes weight distribution for stability during flight. These docking stations can also be used as charging hubs to overcome battery restrictions with hydrogen fuel cell technology. CLOUD includes an OBFP system (Optimal Battery Flight Path) where each drone's flight path is optimized for minimal trips back and forth to charging stations. These charging stations and dock systems are solar powered aiming to have as minimal possible environmental impact as possible.

c. Projected Improvements

Future iterations of the CLOUD drone will concentrate on improving autonomous docking, optimizing electrostatic spraying, increasing flying duration, and improving AI precision. By using real-time machine learning adjustments, the drone will be able to modify its spraying in response to real-time multispectral data, increasing productivity and cutting down on fertilizer waste. While multi-drone coordination will maximize large-scale deployment, a hybrid power system that combines solar-assisted charging and hydrogen fuel cells will increase operational time. Furthermore, biodegradable nanofertilizers, soil health monitoring, and AI-driven sustainability initiatives will improve long-term agricultural profitability and lessen their negative effects on the environment.

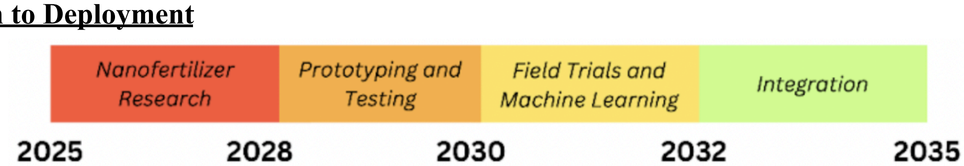
D. Total Cost and Return on Investment

To identify the return on investment (ROI) for the implementation of nanofertilizer drones, one must account for improvements in nanofertilizer versus traditional fertilizer as well as improvements of drones versus tractors.

In the spring of 2024, the average price of fertilizer application across all deployment methods was roughly $160 per acre with the average size of a US farm being 463 acres. This brings the average cost of fertilizer application to $74,080 for an average US farm [44]. However, estimates for fertilizer application via drone vary between $10 to $50 per acre [53]. For an average farm size, nanofertilizer application will cost $13,890. This results in a net surplus of $60,190 per year. The current initial cost of a tractor is roughly $250,000 with a fuel and maintenance of $35 per acre. Maintenance cost estimates are $66,205 for a farm size of 463 acres for traditional fertilizer application techniques. The estimated cost of TITAN is $40,000 with maintenance costs being as low as roughly $1 per acre citing similar drone projects [54]. The yearly estimate for the drone system is $43,215 (40,000 + 5*463). The tradeoff of using a drone versus a tractor is a surplus of $33,000 yearly. All in all, the difference between using nanofertilizers and drones versus the current dispersion system is approximately $93,000 per year.

Currently, the agriculture sector in the US produces about 450 million tons of CO_2 [45]. TITAN's estimated CO_2 emissions are 225 million tons. Tractors produce roughly 40 tons of CO2 a year [46, 47]. On average, a farm has 3 tractors, which results in roughly 120 tons of CO2 generated by a farm from tractors in a given year. TITAN's systems are electric and powered by solar cells. The yearly CO2 output of solar power is much less compared to a tractor. The table below highlights how our system will break even in year 3. Profits will **increase by $93,000 yearly per farm**. If fully adopted, the TITAN system is estimated to reduce US CO2 emissions by 3.6 percent (roughly 230 million metric tons). [See Appendix A]

IV. <u>Path to Deployment</u>



A. Technology Readiness

The mechanical design of both AURAS and CLOUD utilize technology that is available in commercial operations such as the FIXAR 007 drone and existing hexacopters used for liquid deployment [39]. Additionally, nanofertilizer research in recent years has been promising, with small-scale field trials producing exciting findings in crop yield and sustainability.

B. Barrier Analysis

There are several barriers before the proposed system can be implemented. These concerns need to be addressed for successful implementation and future developments on the path to deployment.

1. Drone Regulations

The Federal Aviation Administration (FAA) drone regulations include a weight limitation of 55 pounds including payloads, travel limited to 100 mile-per-hour ground velocity, and a maximum altitude of 400 feet above ground. TITAN operates well under these constraints both in velocity and altitude, and AURAS does not alter the environment. As such, we do not anticipate major barriers due to FAA regulations on drone use. However, geographical restrictions on UAV usage may affect usability in some regions, and regulations on nanofertilizer use present barriers to full integration. Also, the FAA has eased restrictions on drones that must be operated within the "visual line of sight". In 2024, Amazon's delivery drones were given an exemption on the "visual line of sight" ruling paving the way for TITAN to have a similar exemption in the future.

2. Nanofertilizer Research and Regulation

While nanofertilizer technology has been further developed, scaled, and researched in recent years, there still exists little research on the long-term effects on crops, soil, and humans. As such, TITAN's path to deployment begins with research into the long-term effects of nanofertilizers. Research into the long-term effects of agricultural tools can take up to five years when considering the scale of operations and safety measures ensured while analyzing crop and soil health. Another barrier exists with the regulation of nanofertilizers, where TITAN research must confirm nanofertilizer compliance with both the FDA and the TSCA. As such, we first project that this long-term research will combine in the first two steps toward implementation from 2025 to 2030. Following the confirmation of nanofertilizer safety, TITAN will continue with large-scale field trials to ensure soil and human health during aerial deployment across hundreds of acres of farmland. We project that this will take two years referencing similar research trials in recent years. [59]

3. Cost of Manufacturing

The manufacturing of AURAS and CLOUD-compatible drones is costly, especially under strict regulations. In *Thermal Imaging for Your Drone on a Budget*, Duffy highlights the challenge of producing cost-effective autonomous flying drones equipped with high-res and thermal cameras [29]. Therefore, any decrease in the cost of manufacturing nanofertilizers and both drone systems will make the adoption of TITAN more accessible for farms both large and small. The TITAN system would benefit from awards in drone research from the federal government to be able to design AURAS and CLOUD.

C. Conclusion

14 million tons of fertilizer go to waste every year. 18,000 people are killed yearly due to fertilizer pollution in the US. Countless other living organisms die due to algal blooms, hypoxia, soil degradation, and water contamination [48]. One precise drop at a time, TITAN reshapes agriculture—guided by AURAS, delivered by CLOUD, and engineered for a future where every farm thrives without waste.

Appendix A: Return on Investment Table

Timeline	***Start Year 0***	***X years following Initialization***
Cost	*~$ 50,000 yearly (upfront cost of AURAS and CLOUD)*	*Upkeep is close to 0*
Increased Revenue	*0*	*~ $93,000 yearly (Nanofertilizer + drone improvement)*
Bottom Line	*-$50,000*	*-$50,000 + $93,000(x)*
Emissions in the US by Fertilizer/Tractors	*~450,000,000 +120 (per farm) metric tons of CO2*	*220,000,000 metric tons of CO2*

Appendix B: Sources

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